DAMIEN CHEMISTRY OLYMPIAD NOTES

Gaseous State

Question Set with Worked Solutions

Contents: kinetic theory, ideal gases, partial pressure, gas stoichiometry, non-ideal behaviour and Olympiad extensions

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Data and Constants

Unless otherwise stated, assume that gases behave ideally and use

$$R = 8.31 \ \text{kPa} \ \text{dm}^3\text{mol}^{-1} \ \text{K}^{-1}.$$

Temperature must be expressed on the absolute scale:

$$T(\text{K}) = \theta(^\circ\text{C}) + 273.$$

Conceptual Questions

Question 1: Assumptions of the ideal-gas model

Consider the following statements about an ideal gas.
1. Gas particles have negligible volume compared with the volume of the container.
2. Gas particles exert no intermolecular forces on one another.
3. Gas particles lose kinetic energy whenever they collide.
4. Gas pressure arises from collisions of particles with the walls of the container.
5. Gas particles remain stationary between collisions.

Identify the correct statements and explain why each incorrect statement is incorrect.

Answer

The correct statements are **1, 2 and 4**.
Statement 1 is correct because ideal-gas particles are treated as point particles whose own volumes are negligible compared with the volume occupied by the gas.
Statement 2 is correct because the ideal-gas model assumes that there are no attractive or repulsive forces between particles except during collisions.
Statement 3 is incorrect because collisions are assumed to be perfectly elastic. The total kinetic energy of the colliding particles is conserved.
Statement 4 is correct. When gas particles collide with a container wall, their momenta change. The resulting force per unit area produces the gas pressure.
Statement 5 is incorrect because gas particles are in continuous, random motion between collisions.

Question 2: Temperature and molecular speed

Separate samples of neon and argon are maintained at the same temperature. Compare:
1. their average molecular kinetic energies;

Answer

The samples have the same average molecular kinetic energy because average translational kinetic energy depends only on the absolute temperature:

$$E_k = \tfrac{3}{2}kT.$$

Note: This formula relates to the equipartition theorem and degrees of freedom.

Question 3: Why absolute temperature is required

A student substitutes 25 into the ideal-gas equation for a gas at 25 °C. Explain why this is incorrect.

Answer

The ideal-gas equation requires temperature to be measured from absolute zero. Therefore,

$$T = 25 + 273 = 298 \ \text{K}.$$

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Molecular kinetic energy is proportional to absolute temperature. A temperature of 0 °C does not correspond to zero molecular kinetic energy, whereas 0 K is the absolute-zero limit. Using 25 instead of 298 would therefore produce a physically meaningless result.

Question 4: Conditions favouring ideal behaviour

Explain why gases behave most ideally at high temperature and low pressure.

Answer

At high temperature, gas particles possess relatively large kinetic energies. Intermolecular attractions are therefore less significant relative to their kinetic energies.
At low pressure, the particles are far apart. Consequently:

- intermolecular attractions are weak; and
- the particles' actual volumes are negligible compared with the container volume.

These conditions most closely satisfy the assumptions of the ideal-gas model. At low temperature, attractive forces become more important; at high pressure, molecular volume and short-range repulsion become important.

Question 5: Comparing deviations from ideality

Ammonia, methane and helium are examined at the same moderately low temperature and pressure. Which gas would be expected to show the greatest deviation caused by intermolecular attraction? Explain your answer.

Answer

Ammonia would show the greatest deviation caused by intermolecular attraction.
NH_3 molecules are polar and can form hydrogen bonds. Methane is non-polar and experiences only instantaneous dipole-induced dipole forces, while helium atoms are very small and weakly polarisable. The approximate order of attractive forces is

$$NH_3 > CH_4 > He.$$

Stronger attractions reduce the measured pressure because particles are pulled away from the container walls and strike them with slightly less momentum. Thus, ammonia is most likely to have a measured pressure below the ideal-gas prediction.

Question 6: Compressibility factor

The compressibility factor Z is defined as

$$Z = \frac{pV}{nRT}.$$

Explain what is indicated by each of the following:

1. $Z = 1$;
2. $Z < 1$;
3. $Z > 1$.
4. Interesting Extension Question: The Joule-Thomson Effect is the temperature change when a real gas expands from a high pressure area to a low pressure area without any heat exchange. As the molecules spread apart to fill up the larger area, the potential energy increases. By conservation of eneergy, the kinetic energy decreases and temperature falls. In fact, this is the working principle of refrigerators! In order for this principle to work, what value of Z must the real gas have?

Answer

When $Z = 1$, the gas obeys the ideal-gas equation under those conditions.
When $Z < 1$, the gas is more compressible than an ideal gas. Attractive forces dominate, so the measured pressure or volume is lower than the ideal prediction.
When $Z > 1$, the gas is less compressible than an ideal gas. Finite molecular volume and short-range repulsive effects dominate.

Question 7: Adding an inert gas

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A rigid container contains nitrogen at constant temperature. Helium is added without reacting with the nitrogen. Explain what happens to:
1. The partial pressure of nitrogen;
2. The total pressure.
How would the answer differ if the container could expand while the total pressure remained constant?

Answer

For a rigid container at constant temperature,

$$p_{\mathrm{N_2}} = \frac{n_{\mathrm{N_2}} RT}{V}.$$

The amount of nitrogen, temperature and container volume do not change. Therefore, the partial pressure of nitrogen remains unchanged.
Helium contributes an additional partial pressure, so the total pressure increases:

$$p_{\mathrm{total}} = p_{\mathrm{N_2}} + p_{\mathrm{He}}.$$

If the container expands while the total pressure remains constant, adding helium causes the volume to increase. The nitrogen occupies a larger total volume, so its partial pressure decreases. Equivalently,

$$p_{\mathrm{N_2}} = x_{\mathrm{N_2}} p_{\mathrm{total}},$$

and the mole fraction of nitrogen decreases when helium is added.

Question 8: Gas collected over water

A gas is collected by displacement of water. Explain why the measured pressure cannot normally be used directly as the pressure of the collected gas.

Answer

The collected sample contains both the desired gas and water vapour. According to Dalton's law,

$$p_{\mathrm{total}} = p_{\mathrm{gas}} + p_{\mathrm{H_2O}}.$$

Therefore,

$$p_{\mathrm{gas}} = p_{\mathrm{total}} - p_{\mathrm{H_2O}}.$$

The vapour pressure of water depends on temperature and must be subtracted before the ideal-gas equation is applied to the dry gas.

Question 9: Comparing molecular models

Two simplified models are proposed for real gases.
- **Model I:** particles have negligible volume but attract one another.
- **Model II:** particles have finite volume but do not attract one another.

Experimental measurements show that a gas has $Z < 1$ at moderate pressure but $Z > 1$ at very high pressure. Evaluate the two models.

Answer

Model I can explain $Z < 1$ at moderate pressure: attractive forces pull the particles closer together and reduce the pressure relative to the ideal prediction. However, it cannot adequately explain $Z > 1$ at very high pressure because it treats particles as having negligible volume.
Model II can explain $Z > 1$ at very high pressure: finite particle volume reduces the space available for molecular motion, and short-range repulsions become important. However, it cannot explain $Z < 1$ because it contains no attractive forces.
A realistic molecular model must therefore include both intermolecular attractions and finite molecular volume.

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Calculation Questions

Question 10: Determining molar mass

A 0.724 g sample of a volatile substance forms 600 cm^3 of vapour at 101 kPa and 300 K. Determine the molar mass of the substance.

Answer

First convert the volume:

$$600\ \text{cm}^3 = 0.600\ \text{dm}^3.$$

Using $pV = nRT$,

$n = \frac{pV}{RT} = \frac{(101)(0.600)}{(8.31)(300)} = 2.43 \times 10^{-2}$ mol.

Hence,

$$M = \frac{m}{n} = \frac{0.724}{2.43\times10^{-2}} = 29.8\ \text{g/mol}.$$

$$\boxed{M = 29.8\ \text{g/mol}}$$

Question 11: Gas density

Calculate the density of sulfur dioxide at 150 kPa and 350 K. Take

$$M(\text{SO}_2) = 64.1\ \text{g/mol}.$$

Answer

From

$$pV = nRT = \frac{m}{M}RT,$$

we obtain

$$\rho = \frac{m}{V} = \frac{pM}{RT}.$$

Therefore,

$\rho = \frac{(150)(64.1)}{(8.31)(350)} = 3.31\ \text{g/dm}^3$.

$$\boxed{\rho = 3.31\ \text{g/dm}^3}$$

Question 12: Change of gas state

A gas occupies 2.50 dm^3 at 100 kPa and 298 K. Calculate its new volume when the pressure is increased to 160 kPa and the temperature is increased to 350 K.

Answer

Since the amount of gas is unchanged,

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}.$$

Thus,

$V_2 = \frac{p_1V_1T_2}{p_2T_1} = \frac{(100)(2.50)(350)}{(160)(298)} = 1.84\ \text{dm}^3$.

$$\boxed{V_2 = 1.84\ \text{dm}^3}$$

Question 13: Gas stoichiometry

Magnesium reacts with excess hydrochloric acid according to

$$\text{Mg}(s) + 2\ \text{HCl}(aq) \rightarrow \text{MgCl}_2(aq) + \text{H}_2(g).$$

Calculate the volume of hydrogen produced when 0.500 g of magnesium reacts completely at 101 kPa and 298 K. Take $A_{r(\text{Mg})} = 24.3$.

Answer

The amount of magnesium is

$$n(\mathrm{Mg}) = \frac{0.500}{24.3} = 2.06 \times 10^{-2} \ \text{mol}.$$

The equation gives a 1 : 1 mole ratio, so

$$n(\mathrm{H_2}) = 2.06 \times 10^{-2} \ \text{mol}.$$

Using $V = nR\frac{T}{p}$,

$V = \frac{(2.06\times10^{-2})(8.31)(298)}{101} = 0.505 \ \text{dm}^3 = 505 \ \text{cm}^3$.

$$\boxed{V(\mathrm{H_2}) = 505 \ \text{cm}^3}$$

Question 14: Partial pressures from masses

A 10.0 dm^3 container holds 4.00 g of helium and 10.0 g of neon at 300 K. Calculate:

1. the partial pressure of helium;
2. the partial pressure of neon;
3. the total pressure.

Take $M(\mathrm{He}) = 4.00$ g/mol and $M(\mathrm{Ne}) = 20.2$ g/mol.

Answer

For helium,

$$n(\mathrm{He}) = \frac{4.00}{4.00} = 1.00 \ \text{mol},$$

so

$$p_{\mathrm{He}} = \frac{(1.00)(8.31)(300)}{10.0} = 249 \ \text{kPa}.$$

For neon,

$$n(\mathrm{Ne}) = \frac{10.0}{20.2} = 0.495 \ \text{mol},$$

so

$$p_{\mathrm{Ne}} = \frac{(0.495)(8.31)(300)}{10.0} = 123 \ \text{kPa}.$$

By Dalton's law,

$$p_{\text{total}} = p_{\mathrm{He}} + p_{\mathrm{Ne}} = 373 \ \text{kPa}.$$

$$\boxed{p_{\mathrm{He}} = 249 \ \text{kPa}, \quad p_{\mathrm{Ne}} = 123 \ \text{kPa}, \quad p_{\text{total}} = 373 \ \text{kPa}}$$

Question 15: Gas collected over water

A 250 cm^3 sample of gas is collected over water at 298 K. The total pressure is 102.0 kPa, and the vapour pressure of water at this temperature is 3.17 kPa. Calculate the amount of dry gas collected.

Answer

The pressure of the dry gas is

$$p_{\text{gas}} = 102.0 - 3.17 = 98.83 \ \text{kPa}.$$

Also,

$$250 \ \text{cm}^3 = 0.250 \ \text{dm}^3.$$

Therefore,

$n = \frac{pV}{RT} = \frac{(98.83)(0.250)}{(8.31)(298)} = 9.98 \times 10^{-3}$ mol.

$$\boxed{n = 9.98 \times 10^{-3} \ \text{mol}}$$

Question 16: Selective absorption

A 5.00 dm^3 container holds 0.0300 mol of $\mathrm{CO_2}$, 0.0200 mol of $\mathrm{O_2}$, and 0.0500 mol of $\mathrm{N_2}$ at 300 K. All the carbon dioxide is selectively absorbed without changing the temperature or container volume.

Calculate:

1. the final total pressure;
2. the final partial pressure of oxygen.

Answer

After carbon dioxide is removed,

$$n_{\text{final}} = 0.0200 + 0.0500 = 0.0700 \ \text{mol}.$$

Hence,

$$p_{\text{final}} = \frac{n_{\text{final}}RT}{V} = \frac{(0.0700)(8.31)(300)}{5.00} = 34.9 \ \text{kPa}.$$

The oxygen partial pressure is

$$p_{\text{O}_2} = \frac{n_{\text{O}_2}RT}{V} = \frac{(0.0200)(8.31)(300)}{5.00} = 9.97 \ \text{kPa}.$$

$$\boxed{p_{\text{final}} = 34.9 \ \text{kPa}, \quad p_{\text{O}_2} = 9.97 \ \text{kPa}}$$

The oxygen partial pressure is unchanged by removing CO_2 because n_{O_2}, T and V remain unchanged.

Question 17: Adding another gas

A rigid 10.0 dm^3 container initially contains 0.200 mol of nitrogen and 0.100 mol of oxygen at 300 K. A further 0.100 mol of helium is added without changing the temperature.

Calculate:

1. the initial partial pressures of nitrogen and oxygen;
2. the initial total pressure;
3. the partial pressure of each gas after helium is added;
4. the final total pressure.

Answer

Initially,

$$p_{\text{N}_2} = \frac{(0.200)(8.31)(300)}{10.0} = 49.9 \ \text{kPa},$$

$$p_{\text{O}_2} = \frac{(0.100)(8.31)(300)}{10.0} = 24.9 \ \text{kPa}.$$

Therefore,

$$p_{\text{initial}} = 49.9 + 24.9 = 74.8 \ \text{kPa}.$$

After helium is added, the nitrogen and oxygen partial pressures remain unchanged because their amounts, the temperature and the volume remain unchanged. For helium,

$$p_{\text{He}} = \frac{(0.100)(8.31)(300)}{10.0} = 24.9 \ \text{kPa}.$$

Thus,

$$p_{\text{final}} = 49.9 + 24.9 + 24.9 = 99.7 \ \text{kPa}.$$

$$\boxed{p_{\text{N}_2} = 49.9 \ \text{kPa}, \quad p_{\text{O}_2} = 24.9 \ \text{kPa}, \quad p_{\text{He}} = 24.9 \ \text{kPa}, \quad p_{\text{final}} = 99.7 \ \text{kPa}}$$

Question 18: Identifying an unknown gas

A 0.552 g sample of an unknown gas occupies 250 cm^3 at 100 kPa and 300 K. The gas is one of the following:

$$\text{N}_2\text{O}, \quad \text{SO}_2, \quad \text{C}_4\text{H}_8, \quad \text{Cl}_2.$$

Identify the gas.

Answer

Convert the volume:

$$250 \ \text{cm}^3 = 0.250 \ \text{dm}^3.$$

Then,

$$n = \frac{pV}{RT} = \frac{(100)(0.250)}{(8.31)(300)} = 1.00 \times 10^{-2} \ \text{mol}.$$

The experimental molar mass is

$$M = \frac{m}{n} \approx \frac{0.552}{1.00\times10^{-2}} \approx 55.1 \ \text{g/mol}.$$

The candidate molar masses are approximately

Gas	M / g mol^{-1}

N_2O	44.0
SO_2	64.1
C_4H_8	56.1
Cl_2	70.9

The closest value is that of C_4H_8.

The gas is C_4H_8.

Question 19: Reacting and non-reacting gases

A rigid container contains methane, oxygen and nitrogen in the mole ratio

$$CH_4 : O_2 : N_2 = 1.0 : 2.5 : 1.5.$$

The initial total pressure is 105 kPa. The methane is ignited according to

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g).$$

The mixture is then cooled to its original temperature so that all the water formed condenses. Assuming the volume remains constant, calculate:

1. the final total pressure;
2. the final partial pressures of carbon dioxide, oxygen and nitrogen.

Notes

Answer

Take the initial amounts as 1.0 mol CH_4, 2.5 mol O_2, and 1.5 mol N_2. Thus,

$$n_{\text{initial}} = 1.0 + 2.5 + 1.5 = 5.0 \ \text{mol}.$$

One mole of methane consumes 2.0 mol of oxygen. After reaction and condensation of water,

$$n(CO_2) = 1.0, \quad n(O_2) = 0.5, \quad n(N_2) = 1.5.$$

Therefore,

$$n_{\text{final}} = 1.0 + 0.5 + 1.5 = 3.0 \ \text{mol}.$$

At constant temperature and volume, $p \propto n$, so

$$p_{\text{final}} = 105\left(\tfrac{3.0}{5.0}\right) = 63.0 \ \text{kPa}.$$

The final partial pressures are

$$p_{CO_2} = \left(\tfrac{1.0}{3.0}\right)(63.0) = 21.0 \ \text{kPa},$$
$$p_{O_2} = \left(\tfrac{0.5}{3.0}\right)(63.0) = 10.5 \ \text{kPa},$$
$$p_{N_2} = \left(\tfrac{1.5}{3.0}\right)(63.0) = 31.5 \ \text{kPa}.$$

$$p_{\text{final}} = 63.0 \ \text{kPa}, \quad p_{CO_2} = 21.0 \ \text{kPa}, \quad p_{O_2} = 10.5 \ \text{kPa}, \quad p_{N_2} = 31.5 \ \text{kPa}$$

Question 20: Olympiad extension: testing the ideal-gas model

One mole of a real gas is studied at 300 K.

Under condition A:

$$p = 450 \ \text{kPa}, \quad V = 5.00 \ \text{dm}^3.$$

Under condition B:

$$p = 1,000 \ \text{kPa}, \quad V = 3.00 \ \text{dm}^3.$$

Calculate the compressibility factor

$$Z = \frac{pV}{nRT}$$

under each condition. Hence, identify the dominant cause of non-ideal behaviour under each condition.

Answer

Under condition A,

$$Z_A = \frac{(450)(5.00)}{(1.00)(8.31)(300)} = 0.903.$$

Since $Z_A < 1$, attractive intermolecular forces dominate.

Under condition B,

$$Z_B = \frac{(1000)(3.00)}{(1.00)(8.31)(300)} = 1.20.$$

Since $Z_B > 1$, finite molecular volume and short-range repulsive effects dominate.

$$\boxed{Z_A = 0.903 \ \text{(attractions dominate)}, \quad Z_B = 1.20 \ \text{(repulsions and molecular volume dominate)}}$$

This demonstrates why a successful real-gas model must account for both intermolecular attractions and the non-zero volume of gas particles.

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